LESSON PLAN

FOLG (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine Health monitoring by using MCD test
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine Health monitoring by using MCD test
5. Specific Objectives	26.1.1.2 Trainee will be able to explain the MCD EHM Technique

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of MCD test data sheets; Instructor supervision required for data interpretation exercises
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Engine Health Monitoring?
(b) Define the acronym MCD in the context of engine diagnostics.
(c) List three parameters typically monitored during an engine test.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to MCD Test Fundamentals • Understanding EHM Technique Components • Data Acquisition and Processing Methods • Thresholds and Limit Definitions • Interpretation of MCD Results	L & Ds	PPT, IB	15 min
(b)	• Step-by-Step MCD Execution Procedure • Analysis of Deviation from Nominal Values • Correlation with Physical Engine Wear • Reporting Standards for MCD Data • Common Pitfalls in MCD Analysis	L & Ds	PPT, PDF	20 min
(c)	• Integration of MCD with Overall Engine Strategy • Maintenance Decision Making based on MCD • Future Trends in Engine Health Monitoring • Role of Air Agniveer in Data Logging • Summary of Key EHM Concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the MCD test contributes to predictive maintenance.
(b) Describe the relationship between MCD thresholds and engine overhaul schedules.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of MCD testing discussed in this lesson will assist in understanding SOAP Analysis in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine Health monitoring by using SOAP Analysis	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min